

.NET Roadmap 2026

Master the complete .NET ecosystem with the most comprehensive roadmap covering all the latest trends and technologies in 2026.

From ASP.NET Core to cloud-native development, learn exactly what skills **top employers are demanding today**.

What's Inside?

- New Skills to learn in 2026
- .NET, C# and Web Basics
- AI & Essential Tools in 2026
- ASP.NET Core, APIs, Security and ORMs
- Logging, Mapping, Validation and DI
- Architecture, design patterns, and scalable system design
- Databases, caching, messaging, and event-driven systems
- Testing, monitoring, CI/CD, and production deployment
- Cloud platforms, infrastructure as code, and DevOps

Each topic contains the best possible resource to learn the topic (official docs, blogs, videos). For more in-depth training, I have featured the best courses to dive deep into the topics.

Your Instructor: **Anton Martyniuk**, a Microsoft MVP and a Technical Lead

I am helping 80,000+ developers to improve their .NET and Architecture Skills.

I am a founder of Anton DevTips Newsletter, a 3rd biggest .NET Newsletter in the world (22,000+ subscribers).



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New Skills to Learn in 2026

- [New Features in .NET 10 and C# 14](#) (See [10 reasons](#) to upgrade)
- [C# 14 Extension Members](#)
- [Get started with Aspire v13](#) (rebranded from .NET Aspire)
- [Invest Early in Code Quality](#)
- [OpenTelemetry](#) (add observability to every project)
- [GitHub Actions](#) or [Azure DevOps](#)
- Learn a Major Cloud Platform such as [Azure](#), [AWS](#), or [Google Cloud](#)

General Development Skills

- [JSON](#)
- [XML](#)
- [Data Structures & Algorithms](#)
- [Refactoring and Code Quality](#)
- [Clean Code](#)
- [Development Environment & Tooling](#)
- [Security Fundamentals](#)
- [Agile](#)
- [Waterfall](#)
- [CLR](#)
- [BCL](#)
- [.NET CLI](#)
- [NuGet](#)

Web basics

- [HTTP, HTTPS, TLS, DNS](#)
- [Network Layers](#)
- [Web Servers](#)
- [Browser & Client-Side Concepts](#)
- [Web Security Essentials](#)

IDE

- [Visual Studio](#)
- [VS Code](#)
- [JetBrains Rider](#)

AI Tools

- Best reasoning: [ChatGPT](#)
- Best coding model: [Claude Code](#)
- Best IDE integration:
 - [GitHub Copilot](#)
 - [JetBrains AI Assistant](#)
- AI IDE: [Cursor IDE](#)

Best courses:

- [Hands-On: C#](#) (you can execute code and do practical tasks)
- [Refactoring for C# developers](#)
- [JetBrains Rider](#)
- [GitHub Copilot](#)
- [ChatGPT](#) (official, free)
- [Claude Code](#) (official, free)

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Design Principles

- [SOLID Principles](#)
- [DRY Principle](#)
- [KISS Principle](#)
- [YAGNI Principle](#)
- [Domain Driven Design](#)

Design Patterns

- [Creational Patterns](#)
- [Structural Patterns](#)
- [Behavioral Patterns](#)
- [Repository](#)
- [Unit Of Work](#)
- [CORS](#)
- [Saga](#)
- [Event Sourcing](#)
- [Publish-Subscribe](#)

Code design

- [N-Layered Architecture](#)
- [Clean Architecture](#)
- [Vertical Slice Architecture](#)

Best articles:

- [N-Layered vs Clean vs Vertical Slice Architecture: Choosing the Right Approach for .NET Projects in 2026](#)
- [The Best Way To Structure Your .NET Projects with Clean Architecture and Vertical Slices](#)
- [Vertical Slice Architecture: The Best Ways to Structure Your Project](#)
- [Building a Modular Monolith With Vertical Slice Architecture in .NET](#)
- [Refactoring A Modular Monolith Without MediatR in .NET](#)

Software Architecture

- [Monolith Architecture](#)
- [Modular Monolith](#)
- [Microservices](#)

Best courses:

- [SOLID Principles](#)
- [Clean Architecture](#)
- [Modular Monolith](#)
- [Domain-Driven Design: Getting Started](#)
- [Domain-Driven Design: Deep Dive](#)
- [Event-Sourcing](#)
- [Microservices: Getting Started](#)
- [Microservices: Deep Dive](#)
- [Vertical Slice Architecture](#)
- [OpenTelemetry](#)

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ASP.NET Core Basics

- [Routing](#)
- [Logging](#)
- [Middlewares](#)
- [Filters and Attributes](#)
- [Configuration and Options Pattern](#)
- [Authentication and Authorization](#)
- [Error Handling](#)
- [Problem Details](#)
- [HostedService](#)
- [BackgroundService](#)
- [Health Checks](#)
- [Response Compression](#)
- [Rate Limiting](#)
- [API Versioning](#)
- [Controllers](#)
- [Minimal APIs](#)
- [FastEndpoints](#)

REST Fundamentals

- [REST Maturity Model](#)
- [REST Constraints](#)
- [HTTP Methods](#)
- [HTTP Status Codes](#)
- [HATEOAS](#)
- [Filtering](#)
- [Sorting](#)
- [Pagination](#)
- [Data Shaping](#)

Dependency Injection

- [Microsoft DI](#)
- [Scrutor for Decorators](#)

Validation

- [FluentValidation](#)
- [DataAnnotations](#)

Mapping

- [Manual Mapping](#)
- [AutoMapper](#)
- [Mapster](#)
- [Mapperly](#)

Logging

- [Microsoft Logging](#)
- [Serilog](#)
- [NLOG](#)
- [Elastic Search](#)
- [Loki](#)
- [Seq](#)

API Documentation

- [Open API](#)
- [Swagger](#)
- [Scalar](#)

Task Scheduling

- [TickerQ](#)
- [Hangfire](#)
- [Quartz](#)
- [Coravel](#)

Communication

- [REST](#)
- [gRPC](#)
- [GraphQL HotChocolate](#)
- [SOAP](#)
- [HttpClient, HttpClientFactory](#)

Best articles:

- [Best Practices For Building REST APIs](#)
- [90% of APIs are not RESTful. What You're Missing and When It Matters?](#)
- [TOP 15 Mistakes Developers Make When Creating Web APIs](#)
- [Getting Started With HotChocolate GraphQL For Building a Social Media Platform](#)

API SDK

- [Refit](#)
- [RestSharp](#)
- [Polly](#)
- [Microsoft Resilience](#)

Best articles:

- [How To Implement Retries and Resilience Patterns With Polly and Microsoft Resilience](#)

Real-Time Communication

- [WebSockets](#)
- [SignalR](#)
- [HotChocolate Subscriptions](#)

Security

- [CORS](#)
- [Keycloak](#)
- [OpenID Connect](#)
- [Cookies](#)
- [JWT](#)
- [Refresh Tokens](#)
- [Basic Auth](#)
- [Token Auth](#)
- [OAuth 2.0](#)
- [ASP.NET Core Identity](#)

Best articles:

- [Configuring HTTPS Redirection and HSTS in ASP.NET Core](#)
- [Authentication and Authorization Best Practices in ASP.NET Core](#)
- [How to Implement Refresh Tokens and Token Revocation in ASP.NET Core](#)
- [How to Customize ASP.NET Core Identity With EF Core for Your Project Needs](#)

Best courses:

- [REST APIs](#)
- [Minimal APIs](#)
- [Logging in ASP.NET Core](#)
- [Dependency Injection in ASP.NET Core](#)
- [gRPC](#)
- [HotChocolate GraphQL](#)

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SQL Basics

- [SELECT FROM](#)
- [WHERE](#)
- [ORDER BY](#)
- [GROUP BY, HAVING](#)
- [JOINS](#)
- [Database views](#)
- [Stored Procedures](#)
- [Functions](#)
- [Triggers](#)

SQL Databases

- [Postgres](#)
- [MS SQL Server](#)
- [SQLite](#)
- [MySQL](#)
- [Oracle](#)
- [MariaDB](#)

NoSQL Databases

- [MongoDB](#)
- [RavenDB](#)
- [Cassandra](#)
- [Azure Cosmos DB](#)
- [Dynamo DB](#)

ORMs

- [Entity Framework Core](#)
- [Dapper](#)

Best courses:

- [Hands-On: Learn PostgreSQL](#)
- [EF Core](#)

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Caching

- [StackExchange Redis](#)
- [OutputCache](#)
- [HybridCache](#)
- [FusionCache](#)

Event Messaging

- [MediatR](#)
- [MassTransit](#)
- [Rebus](#)
- [Wolverine](#)
- [NServiceBus](#)
- [Dapr](#)

API Gateway

- [Traefik](#)
- [Envoy](#)
- [YARP](#)
- [Nginx](#)
- [Service Discovery](#)

Resilience and Observability

- [Resilient Strategies: Polly](#)
- [Structured logging](#)
- [Distributed Tracing](#)
- [Metrics and Alerts](#)
- [Health Checks](#)
- [Load balancing](#)
- [Rate limiting](#)
- [Seq](#)
- [ELK Stack](#)
- [Jaeger](#)
- [Prometheus](#)
- [Grafana](#)

Testing

Frameworks

- [xUnit](#)
- [TUnit](#)
- [NUnit](#)

Assertions

- [xUnit Assertions](#)
- [Shouldly](#)
- [FluentAssertions \(PAID\)](#)

Integration Testing

- [WebApplicationFactory](#)
- [Test Containers](#)
- [Respawn](#)

Frontend Testing

- [Playwright](#)
- [Selenium](#)

Mocking

- [NSubstitute](#)
- [Moq](#)

Fake Data

- [Bogus](#)
- [AutoFixture](#)

Snapshot Testing

- [Verify](#)

Behaviour Testing

- [ReqNRoll](#)
- [SpecFlow NOT maintained anymore](#)

Performance Testing

- [Benchmarks.NET](#)
- [k6](#)
- [NBomber](#)
- [JMeter](#)

Cloud Providers

Azure

- [CosmosDB](#)
- [Azure Service Bus](#)
- [Azure Event Hubs](#)
- [Application Insights](#)
- [Azure Key Vault](#)
- [Azure Storage](#)
- [Azure Functions](#)

AWS

- [DynamoDB](#)
- [SQS/SNS](#)
- [AWS Kinesis](#)
- [AWS Event Bridge](#)
- [AWS Secrets Manager](#)
- [AWS Lambda](#)
- [AWS S3](#)
- [Google Cloud Platform](#)

CI/CD

- [GitHub Actions](#)
- [Azure DevOps](#)
- [GitLab CI](#)
- [TeamCity](#)
- [Jenkins](#)
- [AWS CodePipeline](#)

Infrastructure as Code

- [Terraform](#)
- [Pulumi](#)

Best courses:

- [Testing with xUnit in C#](#)
- [Integration Testing in ASP.NET Core](#)
- [Docker for developers](#)
- [Docker Compose](#)
- [Kubernetes for developers](#)

Best Cloud courses:

- [Azure for .NET Developers: Getting Started](#)
- [Azure for .NET Developers: Getting Started](#)
- [Cloud Architecture in Azure](#)
- [Serverless with Azure Functions](#)
- [Deploying .NET Applications to Azure](#)
- [Building .NET Applications on AWS](#)

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Bonus 1: EF Core

EF Core Basics

- [DbContext](#)
- [DbSet](#)
- [Connection strings](#)
- [Entities](#)
- [Mapping](#)
- [Migrations](#)
- [Creating new entities](#)
- [Updating entities](#)
- [Deleting entities](#)
- [Filtering Data using Where](#)
- [Data Seeding](#)
- [Getting entity by id](#)
- [DbContext.OnModelCreating](#)
- [Entity Configuration File](#)

Modeling Data

- [Data Annotations](#)
- [Fluent API](#)
- [Properties](#)
- [Indexes](#)
- [Unique constraints](#)
- [Conversions](#)
- [Shadow Properties](#)

Entity Relations

- [One-To-One Mapping](#)
- [One-To-Many Mapping](#)
- [Many-to-Many Mapping](#)

Querying Data

- [Eager Loading](#)
- [Explicit Loading](#)
- [Lazy Loading](#)
- [AsNoTracking](#)
- [Filtering](#)
- [Sorting](#)
- [Pagination](#)
- [Projections](#)
- [Aggregation](#)

Advanced Querying

- [Compiled Queries](#)
- [Split Queries](#)
- [Grouping](#)
- [Aggregation](#)
- [Functions](#)
- [Global Query Filters](#)
- [Change Tracker](#)

Advanced Writing

- [Batch Update](#)
- [Batch Delete](#)
- [Transactions](#)
- [Concurrency: Optimistic Locking](#)
- [Concurrency: Pessimistic Locking](#)
- [Transaction Isolation Levels](#)

Database Migrations

- [Add-Migration](#)
- [Update-Database](#)
- [Database Scaffolding](#)
- [Applying Migrations in Production](#)
- [Creating Migrations for Multiple Databases](#)

Advanced Modeling Techniques

- [Owned Entities](#)
- [Table per Hierarchy \(TPH\)](#)
- [Table per Type \(TPT\)](#)
- [Table per Concrete Class \(TPC\)](#)
- [Keyless Entities](#)
- [Complex Types](#)
- [Value Objects](#)

Advanced Topics

- [Interceptors](#)
- [Temporal Tables](#)
- [Calling database views, functions, stored procedures](#)
- [DbContext Lifetime](#)
- [Multitenancy](#)
- [Audit Logging](#)
- [Working with CosmosDB](#)
- [Working with MongoDB](#)
- [Logging and Monitoring](#)
- [Pooled DbContext](#)
- [JSON columns](#)

Patterns

- [Repository](#)
- [UnitOfWork](#)
- [Mapping DDD entities and value objects in EF Core](#)

Bonus 2: .NET Project Template

Get proven, well-structured template for building maintainable Modular Monolith applications in .NET.

ASP.NET Core Web API, EF Core, PostgreSQL, Authentication and Authorization, OpenTelemetry, FluentValidation, Swagger, Docker, and more.

Architecture: Modular Monolith (you can apply it for classic Monoliths and Microservices)

Code structure: Vertical Slice Architecture

Tech stack: ASP.NET Core 10, EF 10, PostgreSQL, JWT auth, OpenTelemetry

Tests: Unit, Integration, Architecture

Logging: Jaeger, Seq

Deployment: Aspire, Docker Compose

Updated to .NET 10 and added Aspire support:

[.NET Template - Free Download](#)

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